

Caluã de Lacerda Pataca

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Fulbright scholar and Ph.D. candidate at the Rochester Institute of Technology (RIT) with research at the intersection of Human-Computer Interaction (HCI) and Computing Accessibility.

EDUCATION

Rochester Institute of Technology , Rochester, New York	
<i>Ph.D. in Computing and Information Sciences</i>	08/2021 – 05/2025 (EXPECTED)
Advisor: Dr. Matt Huenerfauth / Co-advisor: Dr. Roshan Peiris	
Thesis: Improving the Accessibility of Speech for Deaf and Hard-of-Hearing Individuals Through Affective Captions	
University of Campinas , Campinas, São Paulo, Brazil	
<i>MSc in Electrical and Computer Engineering</i>	08/2018 – 05/2021
Advisor: Dr. Paula Dornhofer Paro Costa	
Thesis: Speech-Modulated Typography	
<i>BA and Licentiate degree in Visual Arts</i>	02/2002 – 07/2007
<i>Certificate program in programming (technical high school)</i>	02/1999 – 12/2002

JOURNAL ARTICLES

[J2] Saad Hassan, **Caluã de Lacerda Pataca***, Laleh Nourian*, Akhter Al Amin*, Diego Navarro, Sooyeon Lee, Alexis Gordon, Matthew Watkins, Garreth Tigwell, and Matt Huenerfauth. (2024) [Exploring the Benefits and Applications of Video-Span Selection and Search for Real-Time Support in Sign Language Video Comprehension among ASL Learners](#) *ACM Transactions on Accessible Computing* [*→Equal Contribution]

[J1] **Caluã de Lacerda Pataca**, and Paula D. P. Costa (2023) [Hidden bawls, whispers, and yelps: can text convey the sound of speech, beyond words?](#) *IEEE Transactions on Affective Computing*.

PEER-REVIEWED CONFERENCE PAPERS

[C7] *Forthcoming* **Caluã de Lacerda Pataca**, Saad Hassan, Lloyd May, Michelle Olson, Toni D'Aurio, Roshan Peiris, and Matt Huenerfauth. (2024) [Tangible Emotions: Multimodal Affective Captioning with Haptics Improves Narrative Engagement for d/Deaf and Hard-of-Hearing Viewers](#).

[C6]  **Caluã de Lacerda Pataca**, Saad Hassan, Nathan Tinker, Roshan Peiris, and Matt Huenerfauth. (2024) [Caption Royale: Exploring the Design Space of Affective Captions from the Perspective of Deaf and Hard-of-Hearing Individuals](#). In *Proceedings of the 2024 ACM CHI Conference on Human Factors in Computing Systems*. [CHI'24] [Accept rate: 26.3%] [Honorable mention]

- [C5]  Saad Hassan, **Caluã de Lacerda Pataca**, Laleh Nourian, Gareth Tigwell, Briana Davis, and Will Wagman. (2024) [Designing and Evaluating an Advanced Dance Video Comprehension Tool with In-situ Move Identification Capabilities](#). In *Proceedings of the 2024 ACM CHI Conference on Human Factors in Computing Systems*. [CHI'24] [Accept rate: 26.3%] [Honorable mention]
- [C4] **Caluã de Lacerda Pataca**, Matthew Watkins, Roshan Peiris, Sooyeon Lee, & Matt Huenerfauth. (2023) [Visualization of Speech Prosody and Emotion in Captions: Accessibility for Deaf and Hard-of-Hearing Users](#). In *Proceedings of the 2023 ACM CHI Conference on Human Factors in Computing Systems*. [CHI'23] [Accept rate: 27.6%]
- [C3]  Saad Hassan*, Akhter Al Amin*, **Caluã de Lacerda Pataca***, Diego Navarro, Alexis Gordon, Sooyeon Lee, & Matt Huenerfauth. (2022) [Support in the Moment: Benefits and use of video-span selection and search for sign-language video comprehension among ASL learners](#). In *Proceedings of the 24th International ACM SIGACCESS Conference on Computers and Accessibility*. [ASSETS'22] [Accept rate: 26.5%] [Best paper nomination]
- [C2] **Caluã de Lacerda Pataca**, and Paula Dornhofer Paro Costa (2020). [Speech modulated typography: towards an affective representation model](#). In *Proceedings of the 25th International Conference on Intelligent User Interfaces*. [IUI'20] [Accept rate: 16.7%]
- [C1] **Caluã de Lacerda Pataca**, and Paula Dornhofer Paro Costa (2019). [Tipografia modulada pela fala: avaliação de um algoritmo de geração de prosódia visual em textos](#). In *Blucher Design Proceedings. 9º Congresso Internacional de Design da Informação*.

POSTER PRESENTATIONS

- [P3] **Caluã de Lacerda Pataca**, Matthew Watkins, Sooyeon Lee, & Matt Huenerfauth. (2022) Developing a captioning style that can help Deaf and Hard-of-hearing individuals identify a speaker's emotions, moods, and emphasis. AI@RIT summit.
- [P2] **Caluã de Lacerda Pataca**, & Matt Huenerfauth. (2022) Visualizing paralinguistic dimensions of speech through typography · 2022 CRA-WP Grad Cohort Workshop for IDEALS
- [P1] **Caluã de Lacerda Pataca**, and Paula Dornhofer Paro Costa. (2019) Tipografia modulada pela fala. In: *Proceedings of the XII EADCA Encontro de Alunos e Docentes do DCA*.

PROFESSIONAL EXPERIENCE

Graduate Research Assistant

Center for Accessibility and Inclusion Research (CAIR)

08/2021 – Present

- With a focus on the perspectives of Deaf and Hard-of-Hearing users, I investigate various forms of captions, including affective, prosodic, and haptic, and how they can be used to enhance the expressive quality of captioned content.
- Plan, execute, and evaluate research investigations utilizing diverse methodologies such as surveys, interviews, and usability studies.
- Design and implement prototypes for research projects, either fully interactive systems (e.g., [C3]), or pre-rendered visualizations of acoustic and ML-based data (e.g., [C1, C4]).
- Author technical papers on HCI and Accessibility findings for conferences and journals.
- Supervisory experience in mentoring and coordinating research assistants, interns, and co-op students, overseeing their work on various projects, including ML-based speech analysis, real-time conferencing applications, and caption design systems.

Founder and Head of Design at Preface

10/2006 – 08/2021

- Designed and coordinated UX, UI, and front-ends in mid- to large-sized projects, while also handling branding-related tasks and other design-related work.
- For the work at Preface, my team and I won multiple design awards and were responsible for the digital platforms of projects of national importance in Brazil, such as the National Olympiad in Brazilian History, in which around 200,000 participants participate each year.

INVITED TALKS & GUEST LECTURES

- 10/24 Experimental Research Design · HCIN-600 Research Methods, RIT
- 09/24 Captions beyond words: Paralinguistic accessibility for Deaf and Hard-of-Hearing individuals · 'Artificial Intelligence for Multimodal Signal Processing' Lab at the University of Campinas School of Electrical and Computer Engineering
- 03/24 Captions beyond words: Paralinguistic accessibility for Deaf and Hard-of-Hearing individuals · Department of Access Services, RIT
- 11/23 Enhancing Affective Captions with Haptic Feedback · Haptics for Inclusion Symposium, University of Borås, Sweden
- 10/23 Captions beyond words: Paralinguistic accessibility for d/Deaf and Hard-of-Hearing individuals · Communication Sciences and Disorders Department, University of Houston
- 03/23 Visualization of Speech Prosody and Emotion in Captions: Accessibility for Deaf and Hard-of-Hearing Users · ISTE-266 Design for Accessibility, RIT
- 10/22 Introduction to experimental design · HCIN-600 Research Methods, RIT
- 09/22 Identifying emotions, moods, and emphasis through captions · Readability Consortium
- 06/13 Design Issues in Web Application and Website Projects · Semana de Atualização em Informática, UTFPR (Ponta Grossa), Brazil, presented together with Lucas Reis

COLLABORATIONS

Outside of collaborating with students and professors at my own university, I have established a global network of collaborators through networking at conferences and symposiums. Some of these collaborations have already borne fruit, and I have papers in submission that stem from work with researchers from Stanford University (US), the University of Toronto (Canada), and the Gwangju Institute of Science and Technology (South Korea). Others are still in their initial stages, but I am hopeful they will be successful in the future. These include research groups from PXL University of Applied Sciences and Arts (Belgium), the University of Lübeck (Germany), and my alma mater, the University of Campinas (Brazil).

PATENTS

- Paula Dornhofer Paro Costa, and Caluã de Lacerda Pataca. Método para composição de closed-captions com parâmetros tipográficos representando visualmente qualidades acústicas da fala (Pending. Process BR-10/2021-016125-6).
- Ricardo Alexandre Da Silva Shimabukuro Victorio, Gustavo Vilela Vargas, Fernando Augusto Chufi, Thiago Yoshihiro Ito Takahashi, Fábio San Juan, Caluã de Lacerda Pataca, José Luis Ponciano Bomfim, Adriana Ramos, Guilherme de Oliveira Feliciano, and Maria Cristina Marcucci Ribeiro (2012). Traceability system applied to farm production activities, industrialization and commercialization of bee products. U.S. Patent Application No. 12/596,612.

SCHOLARSHIPS & AWARDS

- 2021–2026 Fulbright/Capes Full Doctorate Program · *The highly competitive Fulbright Scholarship program supports doctoral study at top U.S. institutions. (\$120,000)*
- Two honorable mentions at CHI'24 · *Top 5% of conference papers.*
- Best paper nomination at ASSETS'22 · *Top 6% of conference papers.*
- 2022 CRA-WP Grad Cohort Workshop for IDEALS · *This competitive program for students underrepresented in the computing field provides mentoring for graduate students across North America.*
- SIGCHI Student Travel Grant (SSTG) to present at the 2020 International Conference on Intelligent User Interfaces at Cagliari, Italy · *Competitive program providing support to students attending SIGCHI-sponsored or co-sponsored conferences.*
- 3 bronze medals, 1 silver medal, and 2 Popular Vote Trophies in 2018, 2019, and 2020 editions of the Brasil Design Award · *The awards were presented at the annual ceremony hosted by the Association for Brazilian Design Companies (ABEDESIGN).*
- Selected project for the 13th Brazilian Graphic Design Biennial · *Prestigious juried exhibition organized by the Brazilian Association of Graphic Designers.*

PROFESSIONAL SERVICE

Associate Chair (AC) for the Late-Breaking Work Track

ACM Conference on Human Factors in Computing Systems (CHI'25) 2025

Web and Graphic Design Chair

ACM SIGACCESS Conference on Computers and Accessibility (ASSETS'25) 2024-25

Technical Paper Reviewer

ACM Conference on Human Factors in Computing Systems (CHI'23, '24 & '25) 2022-present

Late Breaking Work Reviewer

ACM Conference on Human Factors in Computing Systems (CHI'23) 2023

Technical Paper Reviewer

Congresso Nacional de Iniciação Científica em Design (CONGIC) 2021

UNIVERSITY SERVICE

Communications Chair

CAIR, Rochester Institute of Technology 08/2022 – Present

PROFESSIONAL DEVELOPMENT

The Expressive Function of Prosody in Speech

(A função expressiva da prosódia na fala), with Plínio A. Barbora, Ph.D. 2019

Scientific Writing Workshop

(Workshop de Escrita Científica), with Inês Signorini, Ph.D. 2019

Introduction to Computational Design

(Introdução ao Design Computacional), with Guilherme Vieira 2017

Graphic Production

(Produção Gráfica), with Aline Valli 2013

Branding and Culture

(Branding e Cultura), with André Stolarski 2012